Chapter 3
Knowledge Representation & Acquisition

What is Knowledge Representation?

- Way to represent information about the world in a form that a computer system can use.
- Core to intelligent behavior in AI systems.
- Helps machines reason, infer, and make decisions.

Importance of Logic in AI

Logic offers a structured, formal, and precise way to represent knowledge.

Enables:

- Automated reasoning
- Deriving conclusions from known facts
- Verifying knowledge consistency

Types of Logic in AI

Propositional Logic

First-Order Logic (Predicate Logic)

Modal Logic

Temporal Logic

Non-monotonic Logic

Propositional Logic

- **Logical constants**: true, false
- **Propositional symbols**: P, Q, S, ... (**atomic sentences**)
- Wrapping **parentheses**: (…)
- Sentences are combined by **connectives**:

$\wedge$ ...and [conjunction]

$\vee$ ...or [disjunction]

$\Rightarrow$ ...implies [implication / conditional]

$\Leftrightarrow$ ..is equivalent [biconditional]

$\neg$ ...not [negation]

Example of PL Sequence

- P means "It is hot."
- Q means "It is humid."
- R means "It is raining."
- $(P \wedge Q) \rightarrow R$

  "If it is hot and humid, then it is raining"
- $Q \rightarrow P$

  "If it is humid, then it is hot"
- A better way:

  Hot = "It is hot"

  Humid = "It is humid"

  Raining = "It is raining"

Limitation of Propositional Logic

- **Propositional logic has limited expressive power unlike natural language.**
    - Suppose we want to express:

      > "If it rains, the ground gets wet."

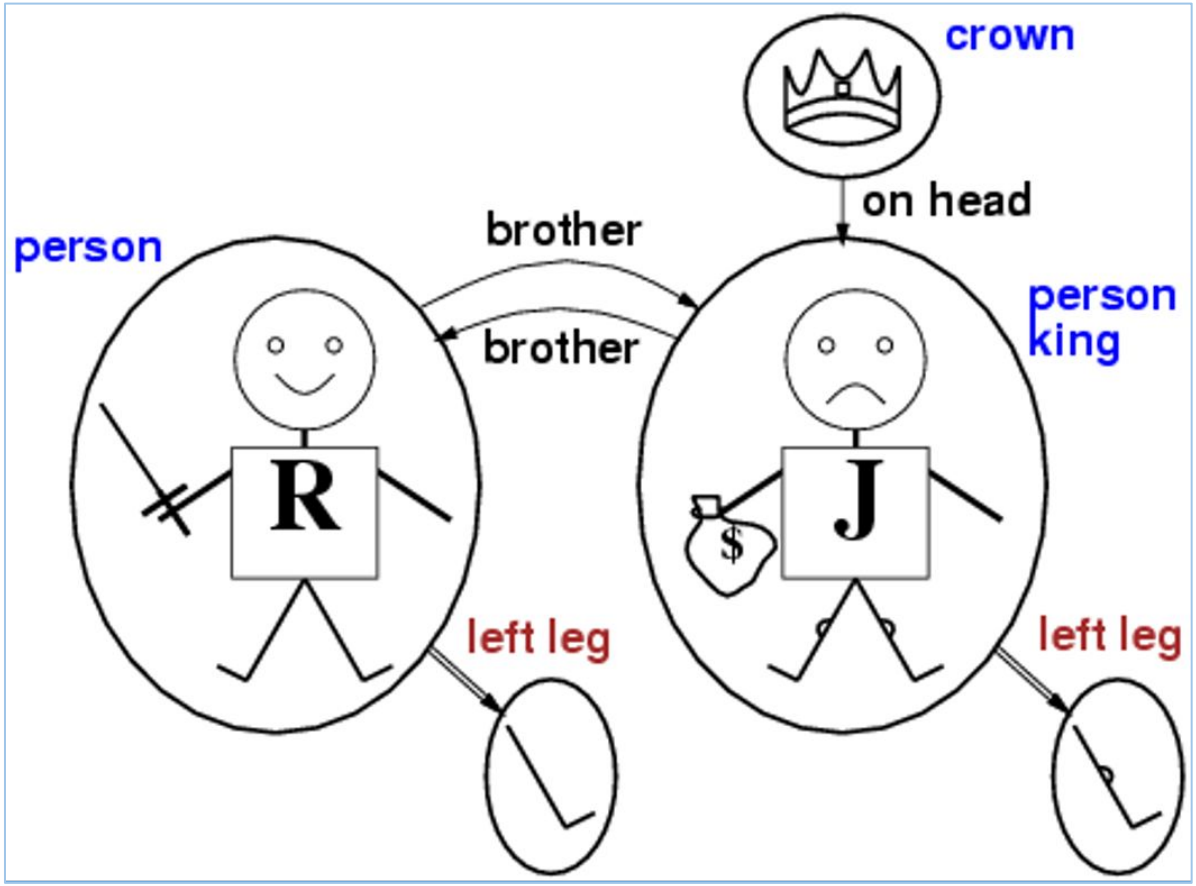
    - **Propositional logic approach:**
        - Let `R` = "It rains"
        - Let `W` = "The ground is wet"
        - Rule: `R ⇒ W` ✅
    - **Now, if we have multiple locations:**
        - `R1` = "It rains in city 1" → `W1` = "Ground wet in city 1`
        - `R2` = "It rains in city 2" → `W2` = "Ground wet in city 2`
    - **Limitation:**
        - For **every city**, we need a **separate proposition**: `R3 ⇒ W3`, `R4 ⇒ W4` ...
        - This quickly becomes **tedious and unscalable**.

Predicate Logic (First Order Logic)

- Propositional logic assumes that the world contains facts.
- First-order logic (like natural language) assumes the world contains
  - Objects: people, houses, numbers, colors, baseball games, wars, …
  - Relations: red, round, prime, brother of, bigger than, part of, comes between,..
  - Functions: father of, best friend, one more than, plus, …

First-Order Logic adds the ability to talk about objects, their properties, and relationships.

First-Order Logic (FOL)



Inference Rules

- The process of deriving **new sentences (facts, conclusions)** that logically follow from a given **Knowledge Base (KB)** written in predicate logic.

- An inference rule is **sound** if every sentence X produced by an inference rule operating on a KB logically follows from the KB. (That is, the inference rule does not create any contradictions).

- An inference rule is **complete** if it is able to produce every expression that logically follows from (is entailed by) the KB. (Note the analogy to complete search algorithms.)

Sound Rules for Inference

- Here are some examples of sound rules of inference
  - *A rule is sound if its conclusion is true whenever the premise is true*
- Each can be shown to be sound using a truth table

RULE	PREMISE	CONCLUSION
Modus Ponens	$A, A \rightarrow B$	B
And Introduction	A, B	$A \wedge B$
And Elimination	$A \wedge B$	A
Double Negation	$\neg\neg A$	A
Unit Resolution	$A \vee B, \neg B$	A
Resolution	$\mathbf{A \vee B, \neg B \vee C}$	$\mathbf{A \vee C}$

Frame in AI

- In AI, frames represent a pivotal concept that helps machines understand and interpret complex real-world scenarios.

- Originating from cognitive science and knowledge representation, frames are utilized to structure information in a way that allows AI systems to reason, infer, and make decisions.

For instance, a "restaurant" frame might include attributes such as "menu," "waitstaff," and "tables," each with its own set of details.

Key Components of Frame

1. Slots

Slots are attributes or properties of a frame. They represent the different aspects or characteristics of the frame's concept.

Example: For a "Person" frame, slots might include:

- **Name:** The individual's name
- **Age:** The individual's age
- **Occupation:** The individual's profession
- **Address:** The individual's home address

Key Components of Frame

2. Facets

Facets provide additional details or constraints for slots, defining acceptable values or specifying how slots should be used.

Example: For the "Age" slot in the "Person" frame:

- **Type:** Integer
- **Range:** 0 to 120
- **Default Value:** 30

Key Components of Frame

3. Default Values

Default values are predefined values assigned to slots if no specific value is provided. They offer a baseline that can be overridden with more specific information.

Example: In a "Car" frame:

- **Make:** Default value could be "Unknown"
- **Model:** Default value could be "Unknown"
- **Year:** Default value could be the current year

Key Components of Frame

4. Procedures

Procedures are methods or functions associated with frames that define how the information within the frame should be processed or utilized.

Example: In an "Account" frame:

- **Procedure:** Calculate Interest - A method to compute interest based on the account balance.

Example of a complete frame

```
+-----------------------------------------------------
|                    Book Frame                      |
+-----------------------------------------------------
| Slots:
| Title: "To Kill a Mockingbird"
| Author: "Harper Lee"
| Publication Year: 1960                             |
| ISBN: "978-0-06-112008-4"
| Genre: "Fiction"                                   |
+-----------------------------------------------------
| Facets:
| Publication Year:
|   - Type: Integer
|   - Range: 1450 to current year
| ISBN:
|   - Format: 13-digit number
+-----------------------------------------------------
| Default Values:                                    |
| Genre: "Unknown" (if not specified)
+-----------------------------------------------------
| Procedures:
| CheckAvailability: Method to check if the book |
|   is currently available in the library.           |
| UpdateRecord: Method to update the book’s          |
|   record when it is borrowed or returned           |
```